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CSE400B Capstone Project

Project Proposal Report

Project Title: *Deep Learning Based Multimodal Emotion Recognition from Audio & Text Modalities*

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1. Introduction

Speech is a method of communication that conveys feelings, thoughts, and emotions through words. There are several methods for showing emotions, including facial expression, speech, text, and physiological signals. Among these modalities, speech (audio) is the most natural and common way to display emotion. This fundamental relationship between speech and emotion leads us to a speech emotion recognition system, where artificial intelligence technology is utilized to identify human emotions articulated in statements.[1]

SER has a wide variety of practical uses, like healthcare, call centers, online streaming media, and smart assistance to assist in keeping track of the home and remote areas,[2] contributing a major impact on human-computer interaction. Human perception of emotions may be biased due to factors like mood, accent, or language. Similarly, SER encounters real-world conditions such as background noise or low-quality audio recording.[3] Despite these challenges, SER has emerged as a significant research domain, enhancing its real-world applications.

Traditional SER systems initially relied on machine learning models with manually selected audio features such as pitch, tone, and spectrograms. However, these approaches often faced limitations due to uncertainty in selecting the most relevant features for accurately identifying different emotions. This field also experienced a transformation with the advancement of deep learning. The research community has shown a preference for deep learning methods because DL models such as CNNs, LSTMs, BiLSTMs, and transformers have shown higher performance in SER by automatically learning complex acoustic and temporal patterns directly from raw or spectrogram-based features.[4] Further progress came through hybrid models (CNN-BiLSTM or attention-enhanced) and transfer learning approaches, especially in real-world conditions.

Despite these advances in audio-based SER, a critical insight appears that emotions are not just encoded in how something is said (audio), but also in what is said (text). This has led to the rise of multimodal SER, which integrates both speech audio and spoken text to enhance emotional understanding. Aligned with direction, this project aims to build a multimodal SER system using the IEMOCAP dataset (containing both audio recordings and their corresponding transcripts). By integrating MFCC and spectrogram-based features for audio and BERT-based embeddings for text, we anticipate that our approach will significantly enhance emotional classification accuracy compared to traditional unimodal methods. Ultimately, this research seeks to contribute to the advancement of SER technologies, paving the way for more effective human-computer interaction across various applications.

1.1 Problem statement

Speech Emotion Recognition (SER) plays an important role in human-computer interaction by accurately identifying emotional states from spoken language. However, most existing SER systems still rely on audio-only input, which often struggle under noisy conditions or speaker variation.[5] Even though deep learning models have improved performance in emotion

recognition,[6] relying exclusively on acoustic features limits emotional understanding. Recent research shows that combining audio with text crucially improve emotion recognition.[7] Despite their promising result, such multimodal approaches are still underutilized in practical applications.

To address these challenges, this project proposes the development of a multimodal SER model on the IEMOCAP dataset and aims to implement a deep learning based fusion model to improve classification accuracy

1.2 Research Questions

The research questions of this research are given below:

- Which feature extraction technique yields the best result in audio emotion classification?
- How can a multimodal SER system be developed by integrating audio and textual features to improve emotion classification accuracy?
- Which architectural approach illustrates superior performance: conventional unimodal SER models, a hybrid model for audio before fusion of text and audio modalities improves emotion classification, or the proposed multimodal fusion?
- How can this model be deployed as a user-friendly application?

1.3 Objectives

To develop a multimodal speech emotion recognition system using the IEMOCAP dataset by integrating audio and text features to enhance emotion classification. The objectives are given below:

- To investigate and evaluate various feature extraction techniques for audio-based emotion recognition and determine the most effective approach for capturing emotional characteristics from speech signals.
- To build a deep learning models for audio features and text embeddings for emotion recognition.
- To implement and facilitate comparison between fusion methods for combining audio and text in a multimodal SER system.
- To create a user-friendly system for real-time emotion prediction.

1.4 Outline

Chapter 1 has the Introduction. It includes an Introduction, Problem statement, Research questions, Objectives, and Expected outcome. Then, Chapter 2 has related Works. It includes the literature survey and a table highlighting performance, models, the dataset used, and the year of publication of the research papers. Finally, Chapter 3 has Methodology. It includes a workflow diagram of the proposed methodology and a brief description of it.

2. Background

Speech Emotion Recognition (SER) has gained increasing attention as a means of improving human-computer interaction by enabling systems to detect emotions from speech. Traditional SER models primarily acknowledged handcrafted audio features such as pitch, tone, and Mel-Frequency Cepstral Coefficients (MFCCs). However, these unimodal approaches often struggled with noisy environments, speaker variability, and confusing expressions. To address these issues, recent studies have found deep learning techniques such as CNNs, LSTMs, and Transformers, which learn complex acoustic and temporal patterns directly from data. Despite these improvements, focusing solely on audio leaves important emotional cues obscured, as the meaning of speech also depends on the textual content. To overcome this shortcoming, our work designs a multimodal SER system based on the IEMOCAP dataset, which fuses frequency and spectrogram-based audio features with BERT-based text embeddings via optimized fusion mechanisms. It's an effort to improve classification accuracy and resilience, making the technology applicable in areas such as healthcare, virtual assistants, and customer service.

The following section provides a concise overview of the literature survey, framing our approach within existing research.

2.1. Related Works

Emotion Recognition does not solely rely on psychological and linguistic features but also considers pitch, tone, gender, and ethnicity. Numerous studies have explored these factors by applying deep learning methods with various models, fusion techniques, and transfer learning frameworks. This section will provide a brief review of the literature relevant to our study.

To show practical uses, this paper [2] introduces a novel intelligent assistant designed to identify human emotions through voice recognition and control peripherals for alert actions using a buzzer and LED alerts, with the capability of detecting seven emotions. The designed system was built on a multilayer neural network, achieving up to 85.12% accuracy while reducing error rate and processing time compared to existing technologies Gaussian Mixture Modeling (GMM) and Support Vector Machine (SVM) Model.

This paper [8] explored whether the fundamental frequency of speech alone is sufficient for emotion recognition. They used the AGH Emotional Speech Corpus, where they examined 280 recordings of 7 emotions from 12 speakers, because of its size, diversity of gender or profession, and availability. The lowest frequency was extracted via Fast Fourier Transformation (FFT) with 20 ms frames (50% overlap) for computational efficiency and noise robustness. Statistical features were derived from low frequency to assess the lowest frequency contours to capture pitch patterns. Five WEKA classifiers were tested for low-frequency discriminatory power across emotion pairs. It was chosen as a computationally cheap, well-established acoustic correlation of emotion, and provides a reliable benchmark for assessing the performance of future multi-feature systems. Achieving up to 89.74% accuracy for two emotions, the findings establish a strong baseline for future emotion recognition systems.

Paper [9]implemented a CNN-based model for real-time emotion recognition on the RAVDESS dataset. The system categorized emotions into four (happy, sad, neutral, and disgust) and achieved an accuracy of 85%. The study focuses on the importance of Discrete Wavelet Transform (DWT) for feature extraction and addresses deployment challenges like background noise and model efficiency in smart assistants and embedded systems. Future directions are harmonious with advancement in multimodal AI.

Paper [10] uses CNN-LSTM and Attention-Enhanced CNN-BiLSTM models for speech emotion recognition on the RAVDESS dataset. The authors transform speech signals into Mel spectrograms and pass them into time-distributed CNN layers to extract spatial features. Then use LSTM layers to model temporal dependencies. The second model upgrades this with bidirectional LSTM and an attention mechanism to focus on emotionally relevant frames. Data augmentation using uniform noise and additive white Gaussian noise (AWGN) was applied to improve generalization. The models achieved 96.5% accuracy for CNN-LSTM and 98.1% for the attention-enhanced version.

In paper,[11] the BanglaSER dataset participants expressed five emotions, which are angry, happy, neutral, sad, and surprise. Human evaluators achieved an accuracy of 80.5%, with the most frequent misclassification observed between surprise and happiness, which accounted for 13.5% of the errors. Experimental results showed that MFCC features achieved 60.86% accuracy, epoch-based features 54.52% while their fusion resulted in an accuracy of 65.93%, reflecting a 5.07% improvement. The remaining 14.52% were below human performance, highlighting the necessity for more advanced feature engineering, data augmentation, and temporal modeling to bridge the gap.

This paper[12] develops an emotion recognition system using the RAVDESS dataset, comparing audio features and deep learning modes. Log-Mel spectrograms are prioritized over MFCCs because they better mimic human auditory processing and retrain critical pitch or energy patterns. A 4-layer 2D CNN was selected for its spatial feature extraction from the spectrogram, while global average pooling handled variable audio lengths to reduce overfitting. Gender specific classification was adopted due to biological acoustic differences between voices. This approach achieved 68% accuracy for RAVDESS and 62% for TESS. Demonstrating that feature engineering outperforms model complexity in speech emotion recognition.

It is important to know the divergence between traditional and deep learning methods for SER before leaping towards a specific path, which has been explored in this paper. The paper[13] traces the evolution of SER algorithms starting from conventional ML methods that require extensive feature engineering and deep domain knowledge to deep learning, emphasizing LSTM-CNN hybrids.

Paper [1] conducted an extensive survey of traditional and deep learning approaches in SER and shows how deep learning approaches outperform classical ML models like SVM, k-NN. Deep learning models like CNN, RNN, LSTM, and GRU were highlighted for their ability to learn complex emotional features automatically. It emphasizes the use of MFCC, Spectrograms, discusses public datasets like RAVDESS, TESS, and EmoDB, and suggests hybrid deep learning models for the future.

To enhance the real-time SER system, this paper[5] used CNN + BiLSTM with data augmentation techniques on RAVDESS, TESS, and EmoDB datasets. Using features like MFCCs, Mel spectrograms, ZCR, RMS, and chroma with data augmentation(noise injection, pitch shifting), the model achieved top accuracies of 100% (TESS), 99.5% (EmoDB), and 90.12% (RAVDESS). This proved that combining spatial and temporal audio features with augmented training superior performance in real-time scenarios and noise-resilient SER systems.

Paper[4] uses a two-stream CNN-BiGRU model on the KEMDy20 dataset and uses both MFCC and Mel-spectrogram features as input. The model allows one stream to process MFCC and the other to process spectrograms and merge them before classification. Achieving 89.71% accuracy, this paper signifies that using multi-feature fusion and recurrent layers improves emotion recognition performance.

Paper[6], [14] proposes a DNN-HMM-based SER system to address the adaptability of speaker variation by combining MFCC with epoch-based features, achieving 65.93% accuracy, an increase rate of 7.13% surpassing existing cutting-edge approaches, suggesting future work with an LSTM network. Their system used a multimodal IEMOCAP dataset to emphasize the importance of deep learning methods in SER.

Hybrid models have shown a significant improvement on both language-dependent and language-independent datasets (RAVDESS, Emo-DB) in this paper[15], [16]. The proposed method of LSTM with the use of Multi-head Attention from Transformers has shown an accuracy of 71.43% on weighted accuracy and 72.50% on unweighted accuracy on the language-independent dataset that was pre-processed into spectrograms from which MFCC were extracted.

Paper[6] introduced a hybrid DL model integrating two parallel CNNs and a Transformer Encoder on the RAVDESS dataset. The authors extract MFCC features from speech signals and treat them as grayscale images, which are input into CNNs for spatial feature learning. A Transformer Encoder is used to capture global temporal dependencies over speech progression. Using Additive White Gaussian Noise (AWGN), data augmentation was performed to expand the dataset. The model achieved 80.46% accuracy, showing improved recognition over the baseline DCNN model (71.61%) and proposing future integration of text-audio features to enhance classification further.

In this paper[15] a lightweight speech emotion recognition system uses a Temporal Convolution Network to process raw speech signals directly. Therefore, it uses a fast network (TCN) on raw sound, because TCN handles long audio patterns better than other networks, enabling real-time use. This means it learns emotions directly in three steps. Then, the final layers combine sound details and timing. Because they focused on speed, it scored 78.34% (IEMOCAP) and 91.61% (EMO-DV) accuracy. Therefore, this helps create practical emotion tools for robots and healthcare.

By highlighting the hindrance in SER development, this study[3] explored TL with embedding models of d-vector, enhanced x and r-vector, with image classification models. As our interest lies with SER, we have focused on the speaker embedding models integrating Multi-Head Attention

pooling and Angular margin SoftMax for improved performance, reflected in the results with an accuracy of 71.08%in unweighted accuracy and 70.97% in weighted accuracy in the enhanced r-vector model. This work has revealed TL to be more effective than data augmentation, significantly outperforming models trained more scratch.

This paper[17] shows that using sound picture tricks with TL on smart networks helps computers learn feelings from speech better when there is little data. So, they hid random spots on sound graphs to stopover memorizing. Inspired by simple emotional trees, they made a step-by-step system using sound and words. So, mixing sound with word understanding gave 74.5% accuracy on IEMOCAP. They say IEMOCAP is now used more than EMO-DB because it's bigger and real-life.

Paper[18] focuses on the theoretical study of multimodal emotion recognition without implementing any actual models. Used MFCC for audio features extraction and CNN for facial features detection. Though no model was evaluated, It highlights the role of accurate feature extraction and the necessity of combining both audio and facial modalities for improving SER systems.

Paper[19] implemented a 3D CNN-based model for speech emotion recognition using EMO-DB and eNTERFACE05 datasets. Reconstructed Phase Space (RPS) analysis is used to transform raw voice signals into 3D tensors, effectively capturing nonlinear emotional features. These tensors are used as input to a 3D CNN model consisting of multiple convolutional, pooling, and dense layers with ReLU and SoftMax activation. The model achieved a high accuracy of 92.02% for EMO-DB and 83.54% for eNTERFACE05 in speaker-independent setups. The study avoids handcrafted features but manifests limitations in computational cost, sensitivity to hyperparameters (τ, d), and dataset validity.

Paper[20] built a system recognizing feelings from both speech and face. Therefore, it used two smart pre-trained networks, which are Audioset and AffectNet. Because they save time by using advanced networks. To focus on key facial parts, they used important maps. To understand changing feelings in videos, they used a special network to track the time. Finally, they mixed the sound and result together. This approach worked well, scoring 80.08% accuracy on the RAVDESS dataset for 8 emotions.

In paper the researchers created MELD, a dataset from Friends dialogues, which addresses the lack of group conversation data for emotion recognition. Also, they used features and text analysis to capture vocal and word cues. Moreover, they chose the DialogueRNN model. Because it tracks individual speakers' states effectively. Therefore, they combined sound and text information. As a result, it improved understanding of short emotional lines where words alone aren't clear. This approach tackled key challenges like context, changing emotions, and speaker interactions.

This[21] research enhances sentiment detection by fusing text and audio data from YouTube reviews. This study achieved superior sentiment classification by integrating CNN and LSTM models to analyze text and audio data. The Bi-LSTM architecture outperformed others, reaching a notable accuracy of 94.97% in distinguishing positive and negative sentiments. This result

shows the model's effectiveness in capturing complex emotional cues, implementing an efficient tool for accurate audience reaction prediction, and enhancing decision-making for content strategies.

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This study[7] introduced novel multimodal models for speech emotion recognition. The audio-only model (ARE) was the weakest performer at 54.6% accuracy. The textual model (TRE) improved results to 63.5%. However, the best model was the Multimodal Dual Recurrent Encoder (MDRE), which used both audio and textual features to achieve an advanced accuracy of 71.8%. This shows that multimodal fusion is essential for accurately classifying emotions into angry, happy, sad, and neutral categories.

M2FNet[23] introduces a multi-modal fusion model that applies text, audio, and visual inputs through parallel attention layers. It achieves best performance, with weighted F1 scores of 66.71% on MELD and 69.86% on IEMOCAP, outperforming existing text-based and multimodal methods in emotion recognition in conversation.

This survey[24] reveals that multimodal emotion recognition significantly outperforms unimodal approaches, with accuracy improvements ranging from 2.35% to 19.73%. Effective models utilize deep learning (DBN, CNN, LSTM, Transformer) for feature extraction and apply fusion methods at the feature level, apply fusion at the decision stage, or combine audio, visual, textual, and physiological signals. These integrated systems achieve higher robustness in classifying both discrete emotions and dimensional states like valence and arousal.

2.2. Table 1: Summary of related works

Ref	**Title**	**Dataset**	**Model**	**Feature**	**Result**	**Emotion Classified**
[1]	Speech Emotion Recognition Systems: A Comprehensive Review of Different Methodologies	Multiple public datasets (RAVDESS EmoDB IEMOCAP Regional language dataset)	1.CNN Based 2. RNN / LSTM 3.Hybrid Model: CNN-LSTM	1.MFCC 2.TEO & TMFCC 3.DWT 4.Hybrid Method: Fusion of MFCC with MS or TEO	Traditional ML: 97% Deep Learning: • 95.89% with 2D CNN-LSTM •92.7% with TMFCC + Deep Neural Network (DNN)	Emotions: Basic (anger, happiness, sadness) and complex (contempt, pride) emotions.

[2]	Multilayer Neural Network Based Speech Emotion Recognition for Smart Assistance	RAVDESS TIMIT Emo-DB	Multilayer Neural Network GMM, SVM, *MLP based on ANN	MFCC, MLP classification	81.02% (RAVDESS) 84.23% (TIMIT) 85.12% (Emo-DB)	Worry, Love, Neutral, Hate, Surprise, Sadness, Happiness (7 emotions)
[3]	Speech Emotion Recognition Using Transfer Learning: Integration of Advanced Speaker Embeddings and Image Recognition Models	IEMOCAP, CREMA-D	Speaker Embedding Models: d-vector, x-vector (enhanced), & r-vector (enhanced)	Pre-trained speaker embeddings	IEMOCAP: Enhanced r-vector: UA- 71.08% WA- 70.97%	Anger, Happiness (happiness & excitement combined), Sadness, and Neutral (4 emotions)
[4]	Speech Emotion Recognition Based on Two-Stream Deep Learning Model Using Korean Audio Information	Korean Speech Emotion Database	1.Bi-LSTM 2.YAMNet 3.Two stream model (Bi-LSTM + YAMNet)	1D Features: MFCC, GTCC 2D Spectrograms: ERB Spectrograms, Log-Mel Spectrogram.	Two-Stream Model: 96% Bi-LSTM: 90.38% YAMNet(CNN based): 94.91%	happy, neutral, angry, sad, chagrin, disgust, fear, and surprise (8 Emotions)
[5]	Real-time speech emotion recognition using deep learning and data augmentation	TESS EmoDB RAVDESS	1.Hybrid model: CNN-BiLSTM 2.MLP 3.CNN	1.MFCCs 2.Mel Spectrogram 3.ZCR 4.RMS 5.Chroma	TESS: 100% EmoDB: 99.5% RAVDESS: 90.12%	TESS: 6 emotions (happy, sad, angry, fearful, disgust, surprise) EmoDB: 7 emotions (anger, boredom, disgust, anxiety/fear, happiness, sadness, neutral) RAVDESS: 8 emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprise)
[6]	Speech Emotion Recognition Algorithm Based on Deep Learning Algorithm Fusion of Temporal and Spatial Features	RAVDESS including AWGN-augmented data	Two parallel CNNs + Transformer Encoder	MFCCs (Mel-frequency cepstral coefficients), input as grayscale images (40×282)	80.46%	neutral, calm, happy, sad, angry, fearful, disgust, surprised (8 emotion)
[7]	Multimodal Speech	IEMOCAP	ARE (Audio Recurrent Encoder), TRE (Text Recurrent	MFCCs, Prosodic Features, Word Embeddings (Text),	Best: 71.8% Worst: 54.6%	Angry, Happy, Sad, Neutral

	Emotion Recognition Using Audio and Text		Encoder), MDRE (Multimodal Dual Recurrent Encoder), MDREA (MDRE with Attention), MDRE-ASR (with ASR transcripts)	MFCCs, Prosodic Features + Word Embeddings, MFCCs, Prosodic Features + Word Embeddings + Attention, MFCCs, Prosodic Features + ASR Word Embeddings		
[8]	Speech Emotion Recognition Based on Voice Fundamental Frequency	AGH Emotional Speech Corpus	K-Nearest Neighbors, Random Forest, Bagging, JRip, Random Subspace Method	Fast Fourier Transform, Statistical feature	RF & Bagging: 89.74%	Anger-Irony, Angry-Fear, Joy-Surprise (3 emotions)
[9]	Real Time Emotion Recognition Using Convolutional Neural Networks	RAVDESS	CNN	DWT	85%	neutral, calm, happy, sad, angry, fearful, disgust, surprised (8 emotion)
[10]	Speech Emotion Recognition: Comparative Analysis of CNN-LSTM and Attention-Enhanced CNN-LSTM Models	RAVDESS	1.CNN-LSTM 2.Attention-enhanced CNN-BiLSTM	Mel Spectrogram (with noise augmentation: AWGN / Uniform)	96.5% (CNN-LSTM) 98.1% (Attention-enhanced CNN-BiLSTM)	Neutral, Calm, Happy, Sad, Angry, Fear, Disgust, Surprise (8 emotion)
[11]	BanglaSER: A speech emotion recognition dataset for Bangla language	BanglaSER	Deep Learning (CNN, LSTMs, GRUs, Transformers)	Amplitude envelope,RMS energy, Zero-crossing rate, Mel-spectrogram, Chromagram, MFCCs, Kurtosis	Best: 65.93%	Angry, Happy, Neutral, Sad, Surprise (5 emotions)
[12]	Emotion Recognition from Speech	RAVDESS	CNN	Log-Mel Spectrogram, MFCCs, pitch, energy, Deltas	-68% of RAVDESS -62% TESS -88% binary emotion classification	Angry, sad, Neutral, Disgust, Happy,Fearful, Surprised (7 emotions)

[15]	TC-Net: A Modest & Lightweight Emotion Recognition System Using Temporal Convolution Network	IEMOCAP, EMO-DB	Temporal Convolution Network, Fully Connected Network, SoftMax classifier	TCN-based SER, Temporal-spatial synergy (TCN, FCN), CNNs	Unweighted Accuracy IEMCAP: 78.34% EMO-DB: 91.61% Weighted Accuracy IEMCAP: 80.84% EMO-DB: 92.31%	Happiness, Anger, Sadness, Neutral, Fear, Boredom, Disgust (7 emotions)
[14]	DNN-HMM-Based Speaker-Adaptive Emotion Recognition Using MFCC and Epoch-Based Features	IEMOCAP	DNN-HMM	MFCC, epoch-based features & MFCC+epoch-based features	(UA considered) MFCC: 60.86% Epoch-based: 54.52% MFCC+Epochbased:65.93%	Angry, Happy, Neutral, and Sad (4 emotions)
[16]	Hybrid LSTM-Transformer Model for Emotion Recognition From Speech Audio Files	RAVDESS Emo-DB language-independent Datasets (combined other two)	*Hybrid LSTM-Transformer (Transform Encoder Layer)	MFCC	75.62% (RAVDESS) 85.55% (Emo-DB) 72.49% (proposed)	Anger, Happiness, Sadness, Fear, Disgust, Neutral, Boredom (7 emotion)
[17]	Improved Speech Emotion Recognition using Transfer Learning and Spectrogram Augmentation	IEMOCAP	pre-trained ResNet34	log-mel spectrograms, spectrogram augmentation,	Exp 1: UA-61.61% WA- 66.02% Exp2: UA-65.56% WA- 65.62% Exp3: UA-64.14% WA- 63.61%	Exp1: Angry, Happy, Neutral, Sad Exp2: Angry, Excited, Neutral, Sad Exp 3: Angry, Excited+Happy, Neutral, Sad
[19]	A 3D Tensor Representation of Speech and a 3D Convolutional Neural Network for Emotion Recognition	EMO-DB (535 samples:German), eNTERFACE05 (1166 samples)	3D CNN	3D Tensors from Reconstructed Phase Space (RPS), τ=17 (EMO-DB), τ=31 (eNTERFACE05), d=3	EMO-DB: (speaker-dependent 92.02%) (speaker-independent 92.48%); eNTERFACE05: (speaker-dependent 83.54%) (speaker-independent 84.96%)	Anger, Boredom, Disgust, Fear, Happiness, Sadness, Neutral (7 emotions)

[25]	Multimodal Emotion Recognition On RAVDESS Dataset Using Transfer Learning	RAVDESS	CNN-14, AlexNet, SVM, Fine-Tuning, Spatial Transformer Network, bi-LSTM, Linear SVM	AlexNet, CNN-14, Linear SVM	CNN-14: 76.58%	Calm, Neutral, Happy, Sad, Angry, Fearful, Disgusted, Surprised (8 emotions)
[21]	MELD: Multimodal Emotional Lines Dataset for Emotion Recognition in Conversations	MELD (Multimodal EmotionLines Dataset)	test-CNN, boLSTM, DialogueRNN, Global GRU, Party GRU, Emotion GRU	Text: 300D GloVe embeddings, 1D-CNN, 100D feature Audio: openSMILE,6373D features, L2 based SVM	Emotion Weighted: 60.25% Sentiment Weighted: 67.56%	Anger, Disgust, Fear, Joy, Sadness, Surprise, Neutral (7 emotions)
[26]	A multimodal hierarchical approach to speech emotion recognition from audio and text	SAVEE RAVDESS IEMOCAP (audio-text multimodal)	ELMo v2 (lexical features), Hierarchical DNN	Prosodic, VQ & Spectral Features, MFCC, Lexical Features (ELMO)	81.7% (SAVEE) 81.2% (RAVDESS) 74.5% (IEMOCAP)	Angry, Happy, Disgust, Sad, Fear, Surprised (6 emotions)
[22]	Maximizing Sentiment Detection through Comprehensive Multimodal Data Fusion: Integrating CNN, RNN, LSTM	IMDb movie reviews & Audio transcripts from YouTube videos	CNN, RNN, LSTM, Bi-LSTM	Multimodal fusion of text features (from CNN) and sequential audio features (from LSTM)	Bi-LSTM achieved the highest accuracy of 94.97% for sentiment classification.	Positive, Negative
[23]	M2FNet: Multi-modal Fusion Network for Emotion Recognition in Conversation	MELD, IEMOCAP	M2FNet with multi-head attention fusion	Text, audio, visual fusion	MELD 66.71%, IEMOCAP 69.86%	Classify emotions such as anger, disgust, sadness, joy, surprise, fear, neutral (MELD); happy, sad, neutral, angry, excited, frustrated (IEMOCAP)

[24]	A review of multimodal emotion recognition from datasets, preprocessing, features, and fusion methods	eNTERFACE05, RML, BAUM-1, MELD, IEMOCAP, SEMAINE, RECOLA, MAHNOB-HCI, DEAP	Feature-level, Decision-level, Hybrid fusion, DBN, CNN, LSTM, Transformer	Handcrafted (LBP, HOG, MFCC, prosodic) and Deep Learning-based features from audio, video, text, and physiological signals	Survey of methods; Multimodal fusion improves accuracy by 2.35% to 19.73% over unimodal	Anger, Disgust, Fear, Happiness, Sadness, Surprise.

3. Proposed Methodology

3.1 Methodology

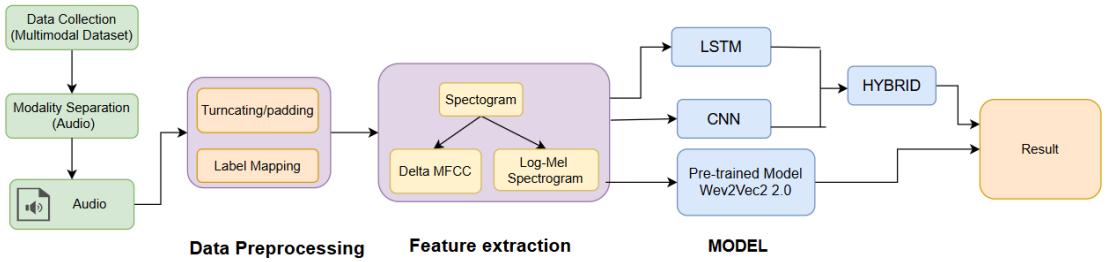


Figure 1: Proposed methodology for Audio Modality

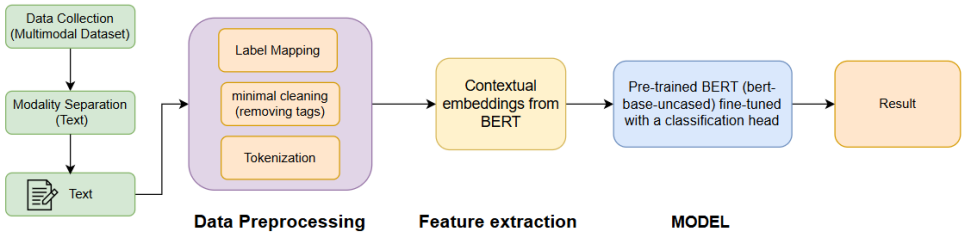


Figure 2: Proposed methodology for Text Modality

1. Data Collection
The IEMOCAP dataset, consisting of audio recordings and text transcripts from scripted and improvised dialogues covering five emotions: happiness, anger, sadness, frustration, and neutrality, will be used in this study.
2. Data Preprocessing
Audio will be resampled, trimmed, and length-standardized using *Librosa*. Text transcripts will be cleaned, lowercased, and tokenized for embedding.
3. Feature Extraction

Delta MFCCs and Spectrograms (Log-mel) will be extracted from audio for speech characteristics, while BERT embeddings will be generated for text to preserve semantic context.

4. Feature Fusion

Early, late, and hybrid fusion strategies will be tested to combine audio and text features, with attention to identifying the most effective integration method.

5. Model Design and Training

Deep learning models with CNN, BiLSTM (for audio), and dense layers (for joint classification) will be developed in PyTorch, trained with categorical cross-entropy loss.

6. Performance Evaluation & Comparison

Models will be evaluated using accuracy, precision, recall, and F1-score, and compared against unimodal and hybrid baselines with cross-validation.

7. Deployment

The final system will be deployed as a user-friendly tool (web or mobile) for real-time audio–text emotion prediction.

3.2 Materials and Devices

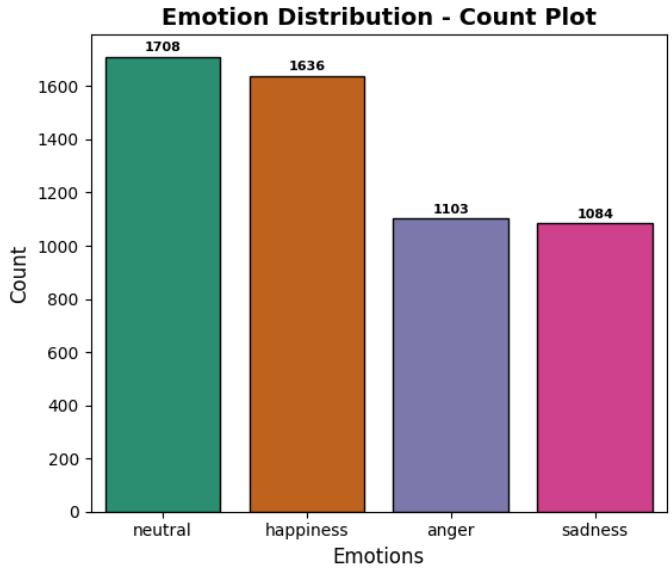
To conduct these experiments, we used various materials and devices:

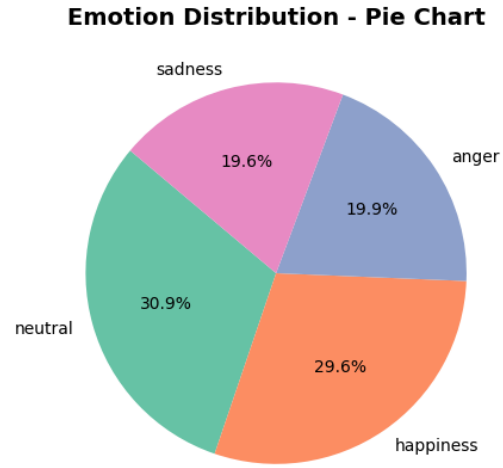
- **Programming Languages:** Python
- **Libraries:** Librosa, SoundFile, Torchaudio, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib, Tqdm, OS, Re, Json, Random, Hugging Face Transformers, Hugging Face Dataset, Warnings
- **Dataset:** Iemocap-full-release
- **Models Used**: LSTM, CNN, BERT, Hybrid (CNN-LSTM), Wav2Vec2 (pre-trained)

3.3 Dataset Description

- The IEMOCAP[27] dataset is a multimodal dataset developed for research enhancing the analysis of emotional states and emotion recognition.
- Approximately 12,000 reactions were taken from five professional actors' recording sessions, totaling the collection.
- It offers rich sources for emotional research by combining many data formats, including audio, video, motion capture (MoCap), and text transcripts.
- The entire speech is marked with overall scores (intensity, excitement, domination) and categorization of emotions (angry, happy, sad, neutral, etc.).
- A minimum of three human individuals examined each observation, verifying the accuracy and consistency of the emotional classification.
- The IEMOCAP dataset originally included 10 emotions, but in our study, we used 4 core categories: *anger*, *happiness* (merged with *excited*), sadness, and *neutral.* This reduction is widely adopted in SER research because the primary emotions have larger sample sizes, while others, like fear, surprise, and disgust, are too sparse for effective training. *Excited* is typically merged with *happiness* since both represent positive, high-arousal states and are often confused when separated. Focusing on these 4 emotions provides a more balanced dataset, enhances classification accuracy, and ensures comparability with prior studies, while still covering the key emotional dimensions relevant for real-world applications.

Below is the emotion distribution in our part of the dataset





4. Design and Implementation

4.1 Audio Modality (Unimodal)

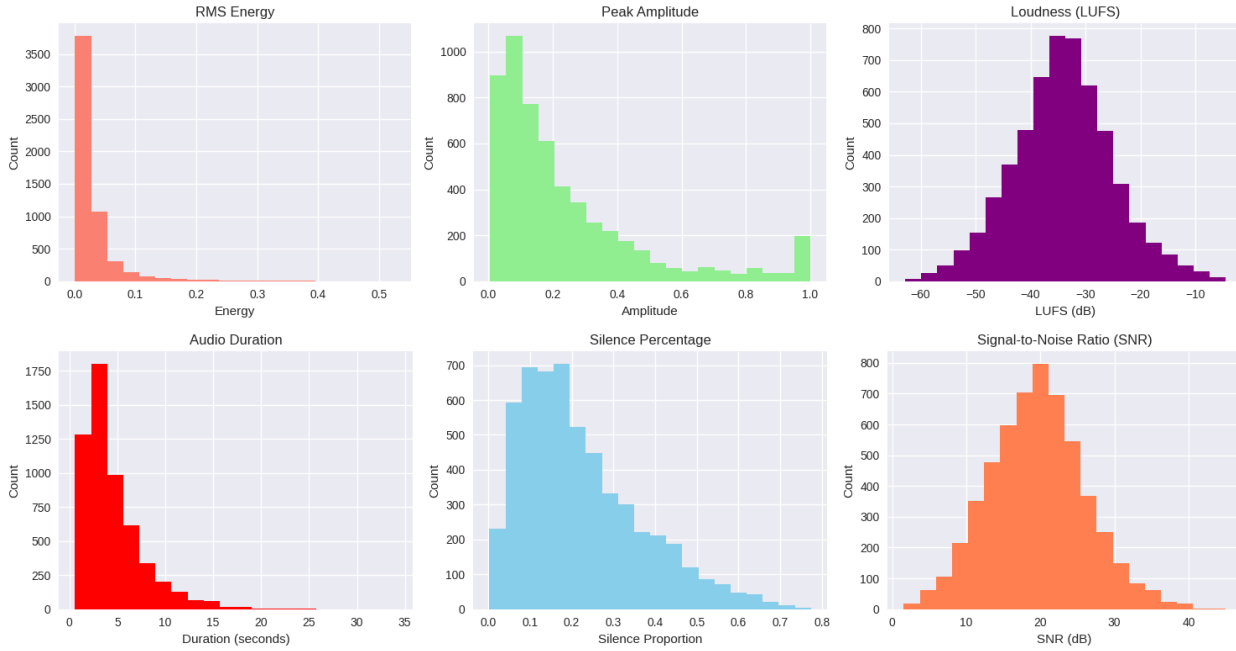
✓ Data Preprocessing & Feature Extraction

Among the work of emotion classification from audio, Mel-frequency cepstral coefficients (MFCCs) and Mel spectrograms and their variants (we have used log-Mel spectrograms) were used the most. Therefore, we have created a pipeline that will extract audios and convert them to MFCCs, which are the audio signal's power spectrum, transformed to mimic how humans perceive sound; along with Log-mel spectrograms - time-frequency representations (2D matrices) that contain temporal information across frames.
We have applied delta and delta-delta features to MFCCs, resulting in a double frame in each feature, allowing the model to learn more about audio signals. As log-mels already contains temporal information, we did not apply delta features to it.

The models used for audio emotion classification process these sequential audio features delta Mel-Frequency Cepstral Coefficients (MFCCs) and Log-Mel spectrograms—and classifies them into four emotion classes: *neutral* (0), *happiness* (1), *anger* (2), and *sadness* (3).

- **Audio Level Statistics**

<u>RMS Energy:</u> Shows a good right-skewed distribution with most samples having low to moderate energy levels, which is typical for speech/audio datasets.
<u>Peak Amplitude:</u> Displays a healthy distribution across the amplitude range with proper normalization (values between 0-1).
<u>Loudness (LUFS):</u> Shows a bell-shaped distribution centered around -30 to -20 dB, which is appropriate for most audio applications.



Audio Duration: Most clips are short (2–6s), suggesting concise emotional expressions. Duration variability has been handled during feature extraction and in model input.
Silence Percentage: Silence ranges from 10% to 50%, potentially diluting emotional content; we have trimmed or padded silence in the model train.
SNR (Signal-to-Noise Ratio): Audio quality is generally good (~20–25 dB SNR), we kept it as it is.

✓ Model Overview

CNN

We have customized a Convolutional Neural Network (CNN) designed for Speech Emotion Recognition (SER) on the IEMOCAP dataset. It processes the two types of input features and follows a progressive down-sampling CNN pipeline with channel attention to improve selective feature learning while reducing the risk of overfitting on the relatively small dataset, which is the probable main reason why we have chosen to go with a customized CNN. Features like MFCCs, and log-Mel features are essentially 2D matrices:

- **Y-axis (rows)** → frequency bins
- **X-axis (columns)** → time frames

This structure is similar to an image (e.g., grayscale images), so CNNs, which are great at image processing—can also process audio features effectively.

LSTM

We have implemented custom Bidirectional Long Short-Term Memory (LSTM) with attention mechanism, and a robust classification head to capture temporal dynamics and focus on emotionally salient parts of the sequence to understand how features evolve over time. Its design captures long-range temporal dependencies and emphasizes the most emotionally informative parts of an utterance, which is an important part of LSTM RNN model structure.

As audio is inherently time-series data and LSTMs are designed for sequence modeling, where the temporal context (what came before and after) affects the current output. Therefore, we have a variant of RNN (LSTM), for our emotion classification that can model how these patterns change over time.

Hybrid CNN-LSTM

To leverage the strengths of both CNNs and RNNs, we have proposed architecture that combines the spatial feature extraction capabilities of Convolutional Neural Networks with the temporal modeling strength of Long Short-Term Memory networks, making it well-suited for speech emotion recognition. We have integrated:

- CNN layers (for local spectral-temporal feature extraction),
- Channel Attention (to emphasize emotion-relevant filters)
- Bidirectional LSTM with Attention (to model long-term temporal dependencies and highlight important time steps).

✓ Architecture Design

CNN

The network has four Convolutional Feature Extraction layers, learning more features with each layer, down-sampling with proper regularization. Inspired by Squeeze-and-Excitation (SE) networks, we have applied global pooling → bottleneck → sigmoid gating to capture emotional salience of spectral regions.

The use of Adaptive Global Average Pooling (GAP) handles variable input sizes, reducing computational time. Two fully connected layers lead to an output layer.

After every convolution, batch normalization stabilizes and accelerates training by normalizing the activations of the previous layer. To map the extracted features to the four emotion classes, classifier of Linear(256→128) → ReLU → Dropout(0.3) → Linear(128→4) was used.

LSTM

The model is structured in three main parts:

<u>(A) Bidirectional LSTM Encoder</u>

enables the model to learn speech dynamics across time — e.g., rising pitch in anger, slower tempo in sadness, capturing temporal dependencies in audio using a stacked LSTM.

<u>(B) Attention Mechanism:</u>

Instead of relying on the final LSTM hidden state (which may lose information from long sequences), attention computes a weighted sum of all time steps to focus selectively on emotionally notable frames. Assign learnable importance weights to each time step in the sequence.

<u>(C) Classification Head:</u>

While not typically used in the recurrent layers, Batch Normalization is highly effective in the final classifier. It accelerates training and provides a slight regularization effect, a vital step for the fully-connected layers that are prone to overfitting. The use of context vector is the "meaning" of the utterance as understood by the model and is what is fed into the classifier.

Their main purpose in our case is to convert the attended sequence representation into class logits.

Hybrid CNN-LSTM

This model is designed for audio classification tasks using 2D time-frequency representations (like spectrograms, MFCCs, etc.).

(A) CNN Feature Extractor

Same convolutional backbone as CNN stated previously has been used to extract hierarchical time-frequency features, down-sampling the input to reduce dimensionality, and applies channel attention to emphasize emotionally relevant channels. The main purpose was to capture local spatial patterns in the frequency-time plane as CNNs are good at learning **local patterns** like harmonics.

Channel attention from the CNN architecture was used to allow the model to reweight channels based on their importance.

(B) Reshaping for Sequential Modeling

The CNN output is reshaped from [batch, channels, height] to [batch, width, channels * height]; where width is the sequence length and channels * height is the feature dimension for the LSTM. This 2D CNN is converted into a sequential format suitable for LSTM input.

We have initialized the LSTM dynamically, as creating the LSTM after the first forward pass allows flexible usage of the model without hardcoding dimensions.

(C) Bidirectional LSTM Component

After the LSTM is dynamically created to match CNN output size, capturing long-term temporal pattern learning, where bidirectionality ensures the model considers past + future context for each frame.

(D) Temporal Attention

To focuses on emotionally significant time steps, aggregating LSTM outputs into a single context vector ensures the model focuses on emotionally informative frames while ignoring silence or neutral content, ending with a fully connected classifier head.

✓ Model Strengths

CNN

The model is comparatively lightweight and can take dynamic input values on a small dataset. Channel Attention boosts performance, and the use of GAP has reduced overfitting, demonstrating a significant advantage over the baseline CNN architecture.

The dynamic input handling for different features is not a standard CNN feature, as most CNNs assume fixed input dimensions. This customization accommodates the dataset's feature variability.

LSTM

A basic LSTM would simply use the final hidden state for classification, assuming all relevant information for the emotion is condensed at the very end of the sequence, which is rarely true.

Emotions are expressed throughout an utterance. This custom design overcomes this fundamental limitation through bidirectionality and the Sequence Attention Mechanism, improving interpretability. The batch normalization and dropout in the classifier form a powerful and robust non-linear mapping from the attended sequence vector to the emotion classes.

Hybrid CNN-LSTM

The CNN is a feature expert, great at recognizing what sounds are present (e.g., a sharp onset, a high-frequency tone). The LSTM+Attention is a temporal expert, great at understanding how the speech unfolds over time.
The custom hybrid architecture combines CNN's ability to extract local time-frequency patterns with LSTM's strength in modeling long-term temporal dependencies, making it suitable for complex audio features like MFCC and Log-Mel spectrograms.
The incorporation of channel attention (from CNN) and temporal attention (from LSTM) focuses on emotionally significant features, addressing the dataset's emotional complexity.
Dynamic input handling is not standard in hybrid CNN-LSTM models, making this model adaptable to the dataset's feature variability.
Dual Attention mechanism, where CNN's feature maps from channel attention and LSTM's output sequence from temporal attention has become one of the greatest strengths of the hybrid model.

- ✓ Training Configuration

All the models were trained on AMP (Asymmetric Multiprocessing) combined with GOU for faster training. For each extracted feature, all the models were run twice to compare the results. Even though the dataset appeared to be imbalanced, no balancing techniques were applied on the dataset.

4.2 Pretrained Model Implementation

To establish a benchmark for performance evaluation, we fine-tuned a pre-trained transformer model - `facebook/wav2vec2-base` model was selected from the Hugging Face ecosystem for this purpose.
The architecture of Wav2Vec2 is based on a self-supervised learning paradigm, where the model learns rich speech representations directly from raw audio waveforms. As a result, no manual feature extraction is required; instead, the raw audio signals extracted from the dataframe are directly provided as input to the model.

- ✓ Dataset Preparation

The dataset was split into training and validation sets. A custom PyTorch class handled audio loading, similar to baseline models, and preprocessing with Wav2Vec2Processor, while bidirectional label mappings ensured smooth classification with HuggingFace's Trainer.

- ✓ Model Configuration

The model was fine-tuned with a classification head adapted for our specific multi-class classification task. The lower-level convolutional feature extractor was frozen, ensuring that only the higher transformer layers and classification head were updated during training. This reduced computational complexity while leveraging strong pre-trained representations. Explicit `id2label` and `label2id` dictionaries were provided to ensure consistent mapping between logits and human-readable emotion labels throughout the training and inference pipeline.

✓ Training Setup

Training was conducted with HuggingFace's Trainer API. A `DataCollatorWithPadding` ensured uniform input lengths across batches by applying dynamic padding. The following hyperparameters were adopted:

Epoch = 8 Batch Size: 4 (train/eval) Weight Decay: 0.01	Learning Rate: 2e-5 (Adam optimizer implied) Logging: Every 10 steps for progress tracking	Evaluation & Saving: Performed at each epoch, retaining the best model based on validation accuracy	**Metrics:** accuracy, macro-averaged F1-score, precision, and recall

This fine-tuned Wav2Vec2 model serves as our high-performance benchmark. We expect this model to significantly outperform our custom CNN, LSTM, and CNN-LSTM models, which are trained from scratch on extracted features (e.g., MFCCs, Log-Mel Spectogram). The comparison will highlight the performance gap between traditional deep learning approaches and modern transfer learning with large pre-trained models in the domain of SER. Even though Wav2Vec2 benefits from large-scale pretraining and strong feature extraction, the custom models are designed to explore lightweight and task-specific alternatives, providing a direct comparison between pre-trained model and deep learning techniques.

4.3 Text Modality (Unimodality)

✓ System Overview

The text-only branch of our multimodal Speech Emotion Recognition (SER) system focuses on building a BERT-based classifier using the IEMOCAP transcripts. Utterances were extracted and mapped into four emotion categories: anger, happiness (with excited merge), sadness, and neutral. Minimal cleaning was applied, including removal of tags such as [LAUGH] and normalization of spacing and punctuation, while preserving casing and semantic cues important for emotion detection. To address class imbalance, the model was trained with class-weighted cross-entropy and evaluated using a session-wise cross-validation strategy, where each fold held out one session for testing to ensure speaker-independent evaluation.

✓ Implementation Pipeline

The implementation was carried out in a structured notebook, following a sequence of stages from data preparation to model evaluation.

✓ Data Acquisition and Validation:

The IEMOCAP dataset was imported into the Kaggle environment and its integrity verified. The directory tree was checked to confirm the presence of all required subfolders (Session1–5, EmoEvaluation, Transcriptions), ensuring early detection of missing or corrupted files.

✓ Data Parsing and Preprocessing:

Custom parsing functions were implemented to align emotion labels with textual transcripts, which involved extracting data from the dataset's complex file structure. A mapping scheme

consolidated related labels (e.g., mapping excited to happiness) and filtered out non-relevant categories, yielding a standardized four-class dataset. The parsed data was merged into a structured Pandas DataFrame, which consists of session, dialogue, utterance, text, and emotion labels.

- ✓ Exploratory Data Analysis (EDA):

EDA was conducted to guide preprocessing and model design. Class distribution visualizations confirmed significant imbalance across emotion categories, while text length analysis determined an optimal maximum sequence length of 128 tokens, covering over 99% of utterances.

- ✓ Text Preprocessing and Feature Engineering:

A lightweight cleaning function prepared text for BERT, normalizing whitespace, compressing repeated punctuation, and removing non-lexical markers (e.g., [LAUGH], [COUGH]). Emotion labels were encoded into integers, and mapping dictionaries (id2label, label2id) were created for model compatibility.

- ✓ Model Design and Training Setup:

The model architecture used a Hugging Face Transformers BertForSequenceClassification and was initialized with the pretrained bert-base-uncased checkpoint. A custom WeightedTrainer subclass was created to add computed class weights into the CrossEntropyLoss function to create a fairer treatment of the minority classes to help overcome class imbalance. To ensure full reproducibility, the states of all randoms were seeded and fixed for NumPy, PyTorch and the Python random state.

- ✓ Model Training and Evaluation:

A rigorous leave-one-session-out cross-validation strategy was employed to assess generalization. For each fold, the model was trained on four sessions and validated on 15% of that training set, while the remaining session was reserved for testing. Training was performed for four epochs using the AdamW optimizer with a learning rate of 2e-5. Performance was tracked via custom callbacks, logging training and validation losses as well as macro-F1 scores at each epoch. Evaluation prioritized the macro-F1 score, supplemented with accuracy and detailed classification reports.

- ✓ Results Analysis and Visualization

After cross-validation, results were aggregated across all folds. Training and validation curves were plotted to detect underfitting or overfitting. A comprehensive classification report and confusion matrix were generated by concatenating predictions from all test folds, providing a holistic view of system performance.

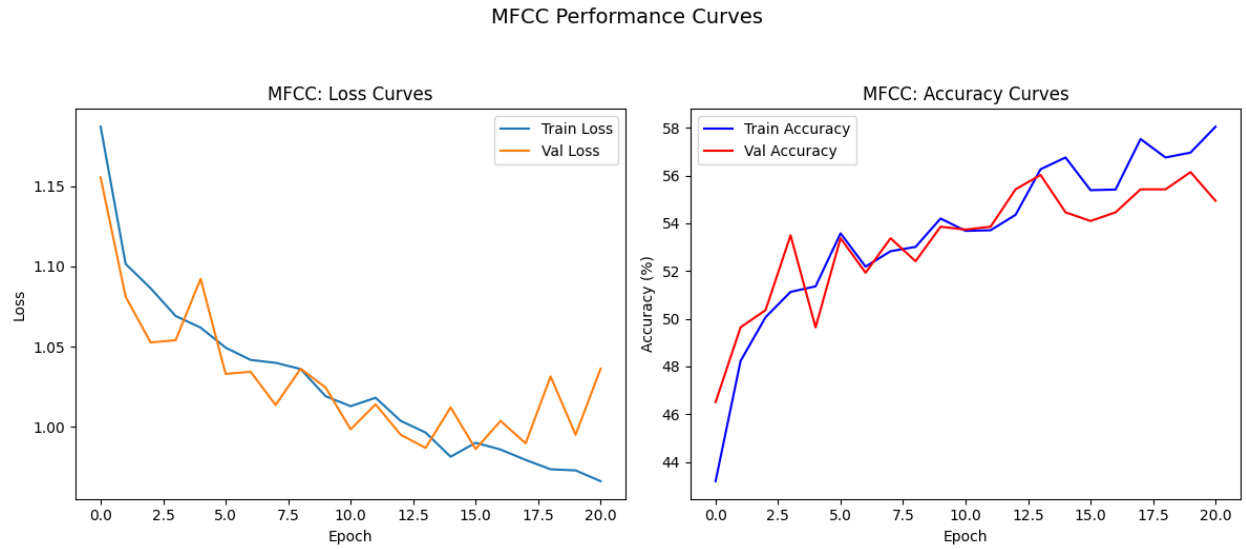
5. Result Analysis

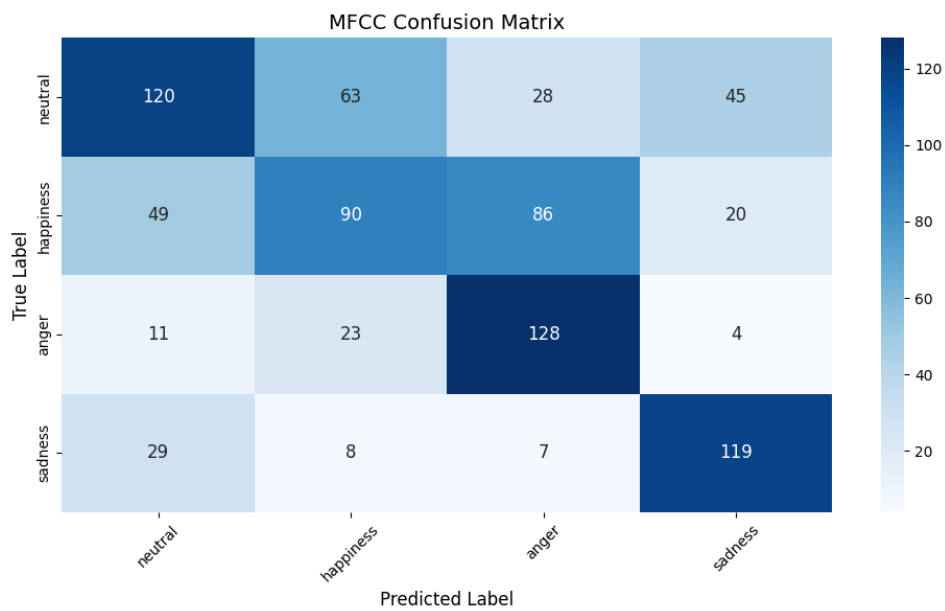
5.1 CNN

On delta MFCCs feature the CNN model performed well compared to the log-mel spectrogram feature, suggesting that the temporal changes in the sound's sound character (captured by Delta MFCCs) were more discriminative for this specific task than the static spectral energy distribution (captured by Log-Mel Spectrograms).

MFCC

Both training and validation loss decrease consistently, while training and validation accuracy improve steadily. The model trained on MFCC features shows a steady improvement in both training and validation performance over 20 epochs. The validation loss fluctuates slightly but follows a general downward trend, suggesting the model is not overfitting.





From the confusion matrix, it reveals that the model performs best on anger and sadness. Neutral and happiness classes show more confusion, especially:

Neutral is often misclassified as happiness (63 times).

Happiness is frequently confused with anger (86 times).

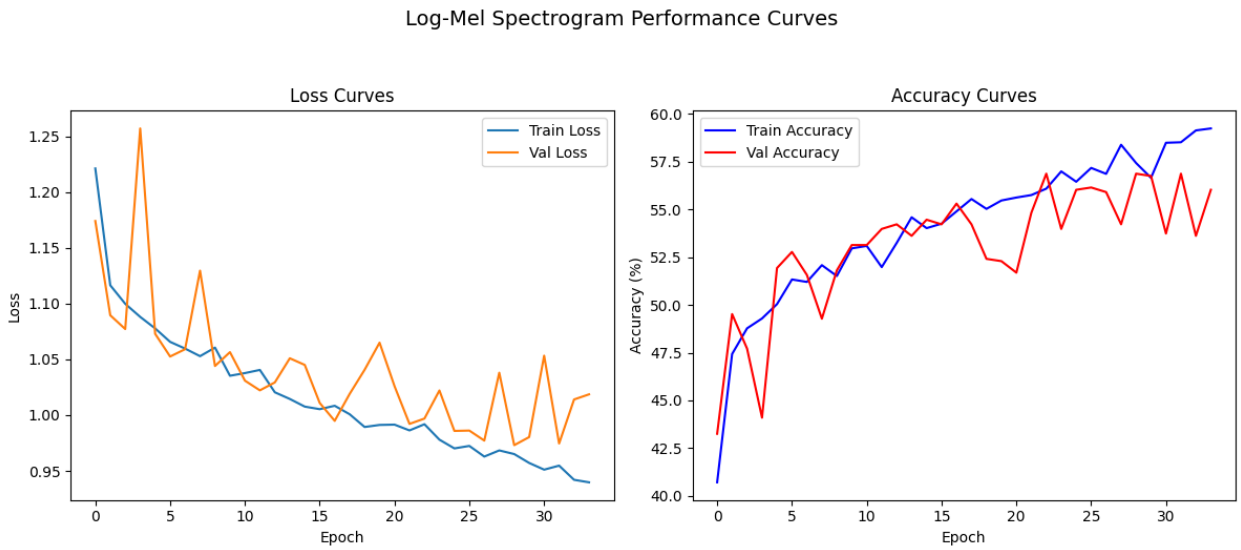
This suggests that the model struggles to clearly separate positive emotions (neutral vs happiness) and may confuse them with more intense expressions like anger.

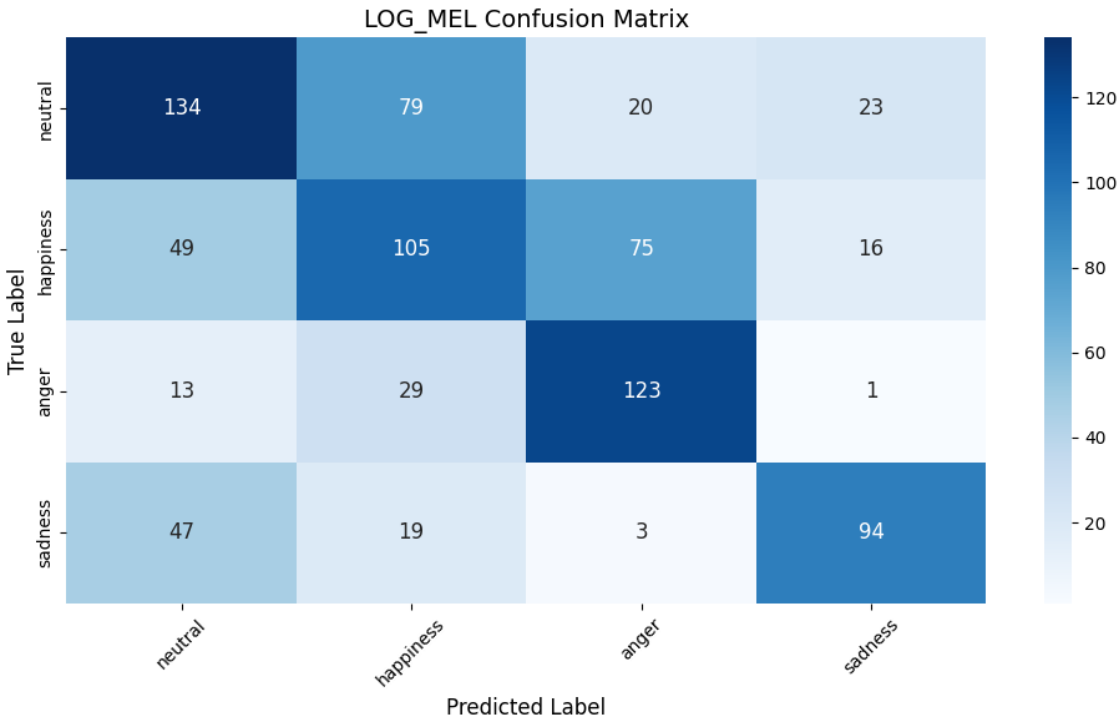
Overall, while the model is effective in detecting anger and sadness, further improvements could focus on better distinguishing between neutral and happiness.

Log-Mel Spectrogram

Both training and validation loss decrease, though validation loss is more unstable, hinting at sensitivity to overfitting or noise. Training accuracy rises steadily to nearly 60%, outperforming the MFCC model, while validation accuracy plateaus at 55–57% and fluctuates after epoch 20, suggesting overfitting.

The Log-Mel model achieves **higher training and validation accuracy** compared to the MFCC model (peaking around 60% vs. 58% for MFCC).





The CNN model shows improved classification performance across most emotions - *neutral* and *happiness* classes are predicted more accurately than in the MFCC model. *Anger* maintains

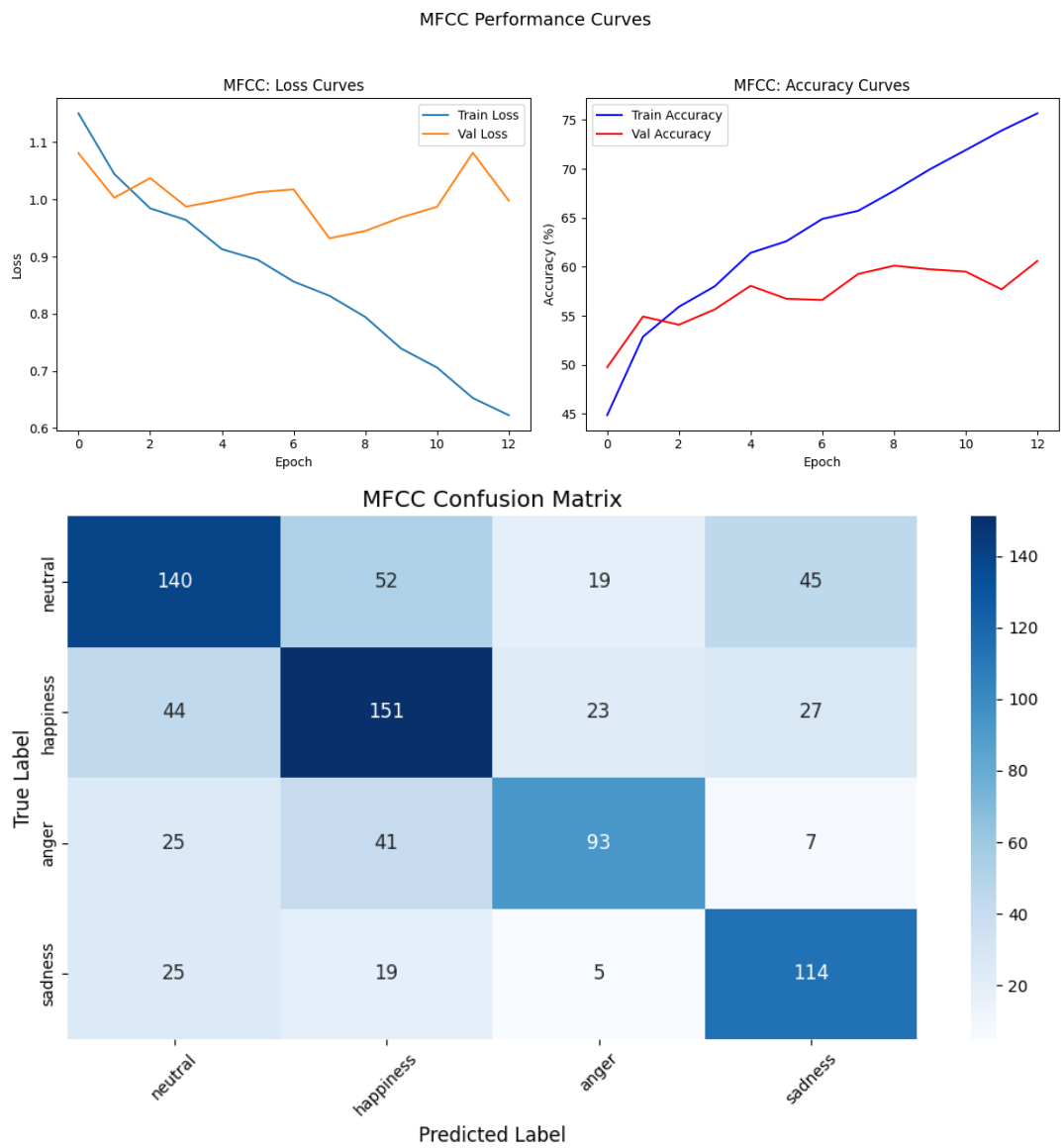
strong performance, though slightly lower than MFCC's. *Sadness* accuracy drops a bit, and it's frequently confused with *neutral*.

5.2 LSTM

In temporal analysis of the LSTM model, MFCC features achieved a peak **accuracy** of **60%**, with an unweighted average recall (**UAR**) of **60.57%**. These results demonstrate a clear improvement over the **CNN**-based model, which achieved an **accuracy** of **55.06%** and a **UAR** of **58.43%**.

MFCC

The LSTM model trained on MFCC features fits the training data well but generalizes poorly compared to CNNs, as shown by steadily decreasing training loss alongside plateauing or rising validation loss.

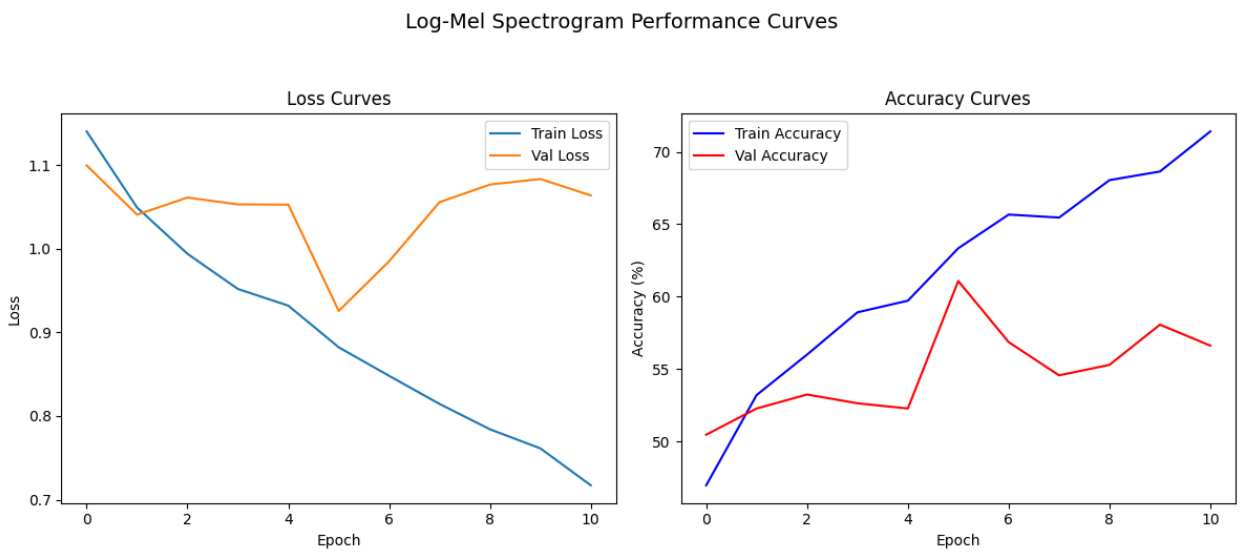


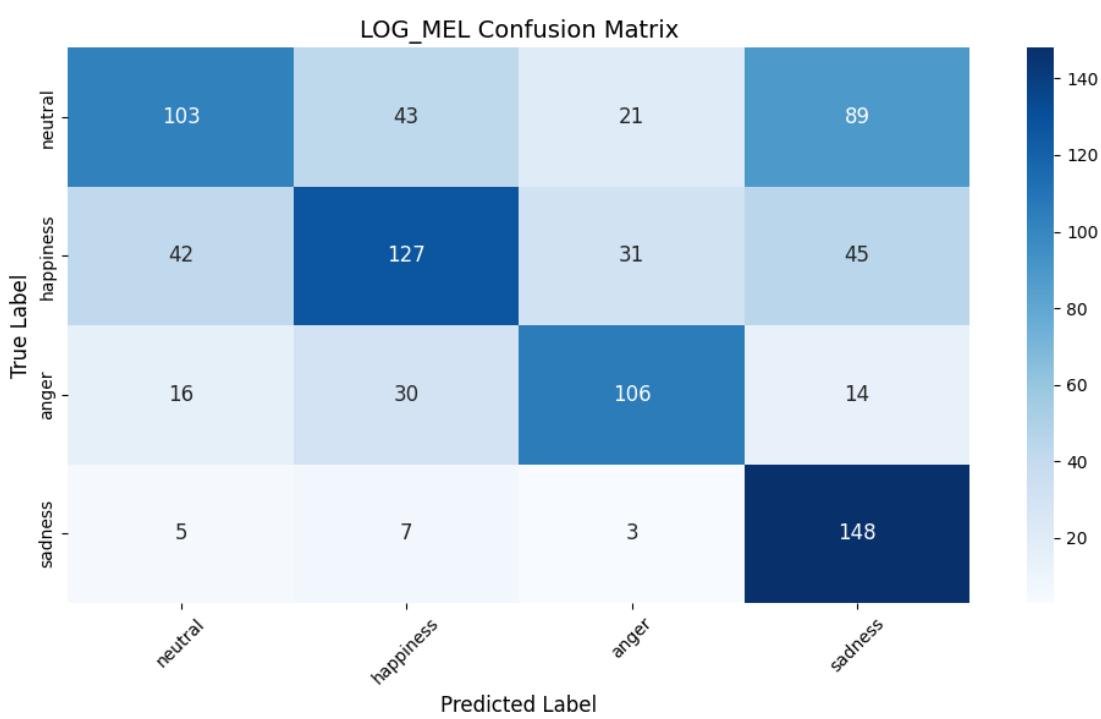
The LSTM model shows generally stronger class separation, especially for *happiness* and *neutral*, which was an issue in the CNN model. *Neutral* and *sadness* are also well-classified; *anger* is the weakest among the four.

Log-Mel Spectrogram

The performance of the log-mel feature on the LSTM model decreased compared to MFCC, but it performed well, similar to MFCC, which outperformed CNNs.

The confusion is log-mel spectrogram appears to be a clearer classification than CNN's log-mel feature classification over the confusion matrix, stating that LSTM performed better overall than CNN.



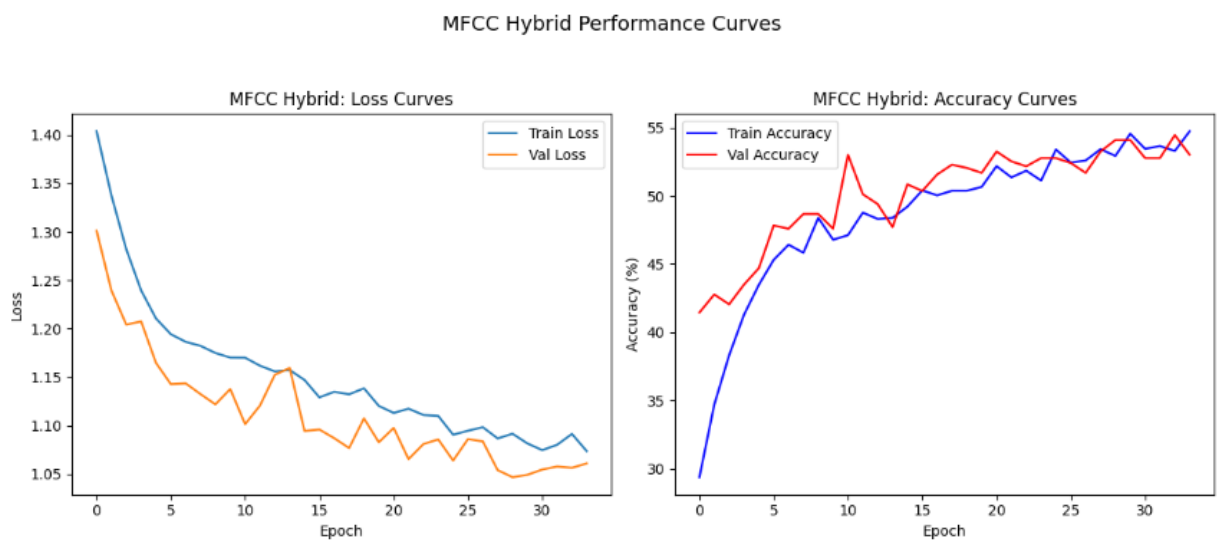


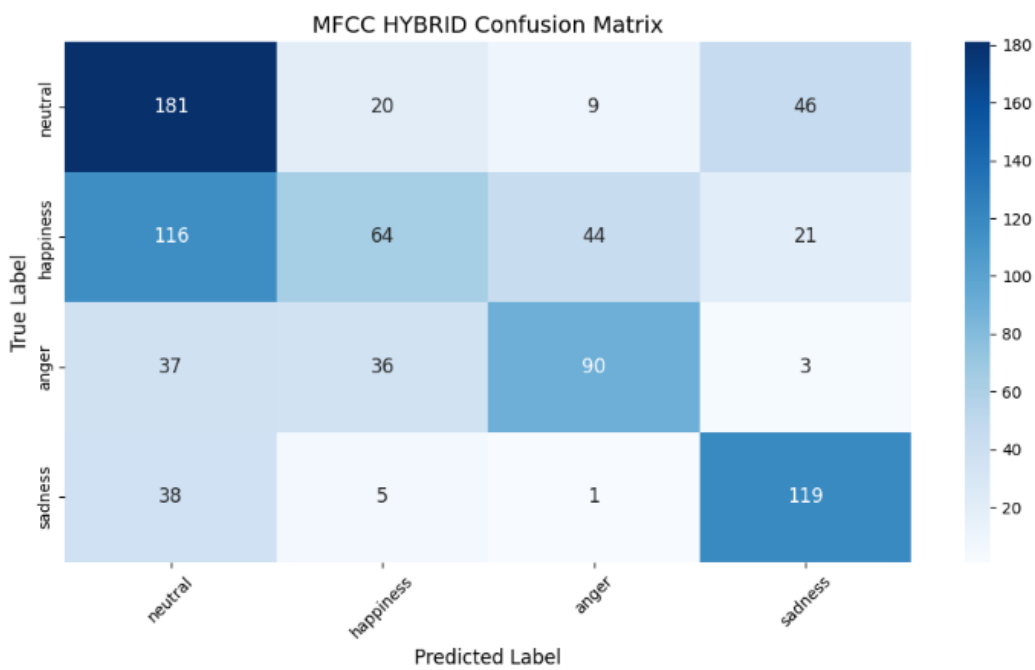
5.3 Hybrid CNN-LSTM

Although the CNN-LSTM hybrid was expected to outperform standalone models by combining spatial and temporal feature learning, it instead underperformed with log-Mel spectrograms, showing significant drops in both UAR and accuracy, suggesting ineffective integration of the two architectures.

- **MFCC**

Loss Curves: Both training and validation losses decrease consistently, with validation loss staying lower than training loss, indicating strong generalization and no overfitting — a rare but desirable outcome in deep learning.
Accuracy Curves: Training and validation accuracy plateau around 55% and track closely, indicating stable learning and generalizable pattern capture without overfitting.

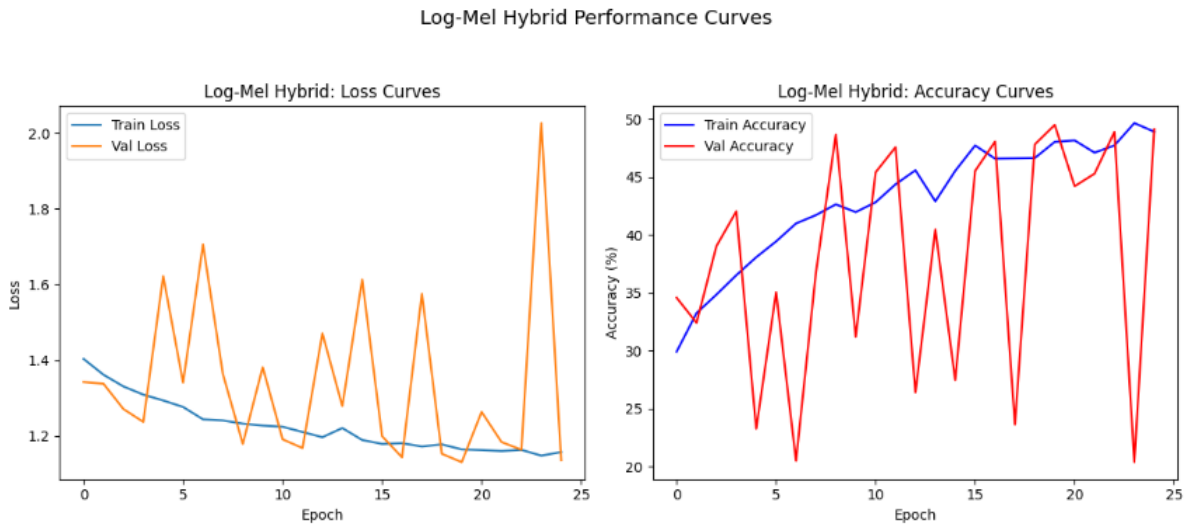




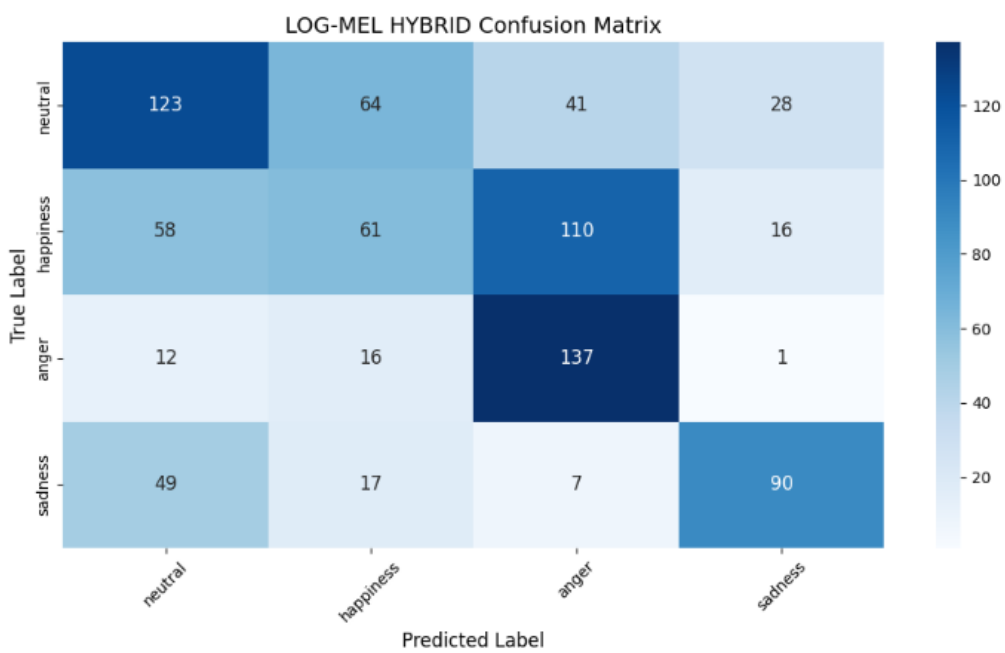
The Hybrid CNN-LSTM model using MFCCs shows excellent classification of *neutral* and *sadness*, moderate performance on *anger*, and consistent difficulty with *happiness*, which is a common challenge across all our tested architectures.

Log-Mel Spectrogram

Compared to the MFCC-based hybrid model, this configuration of the log-mel feature appears unstable and less effective.



Loss Curves: While training loss decreases smoothly, the highly erratic validation loss with sharp spikes—especially near the final epoch—signals instability and poor generalization.
Accuracy Curves: Training accuracy gradually reaches \~48–50%, but highly unstable validation accuracy (25–50%) indicates poor generalization and sensitivity to input variation or class imbalance.



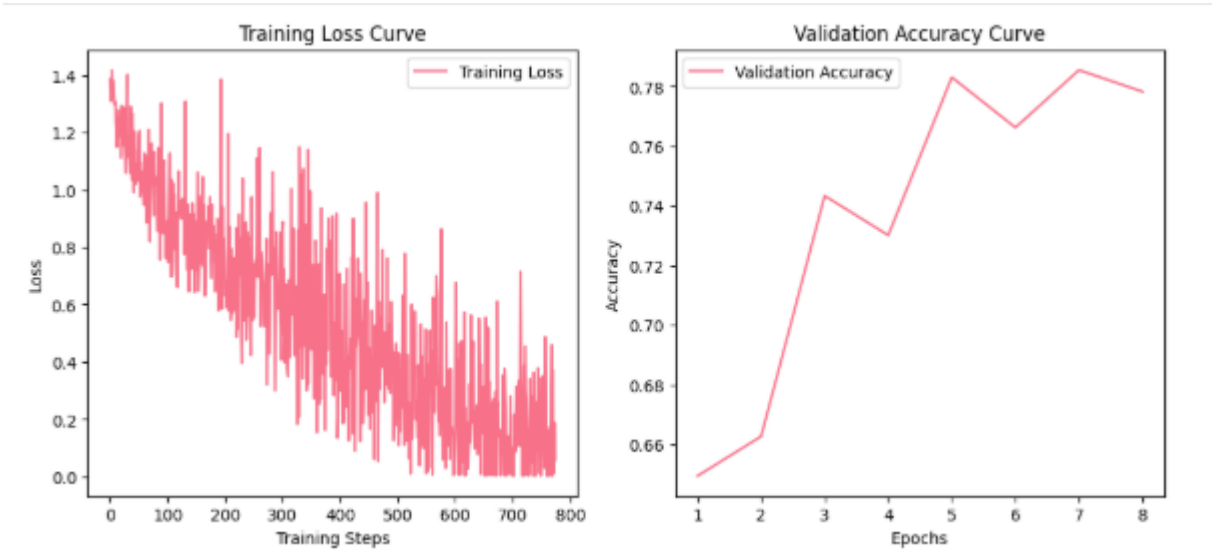
Despite unstable validation curves, the confusion matrix suggests that the model still performs reasonably well in classifying certain emotions, especially *anger*. *Neutral* suffers from major confusion with happiness (64) and anger (41), highlighting ongoing difficulty in separating low-arousal emotions.
While the validation performance of the Log-Mel Hybrid model is unstable, the confusion matrix reveals that it still achieves high accuracy in identifying certain emotions, particularly those with distinct temporal dynamics (e.g., anger).

5.4 Pre-trained Model (Wav2Vec2)

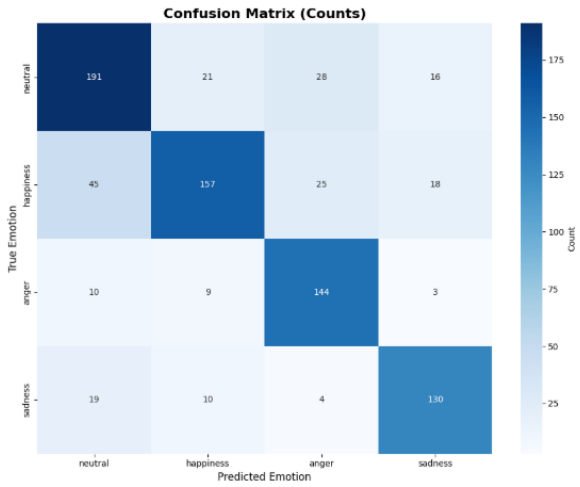
The pretrained model significantly outperforms all previous models—achieving the highest accuracy (74.94%) and UAR (76.30%)—demonstrating its strong ability to generalize across emotional categories.

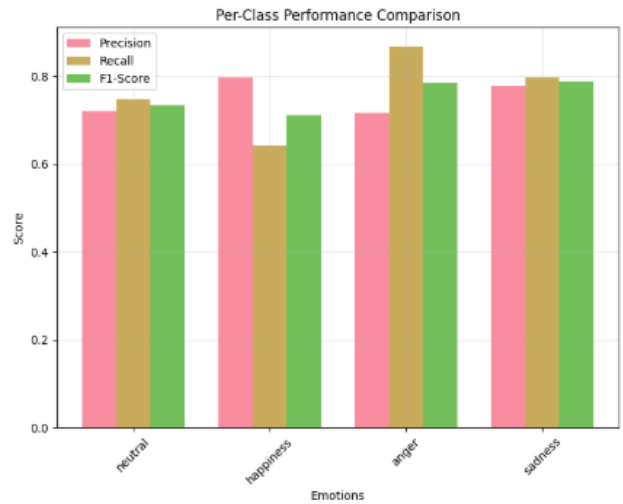
Training Loss Curve: Shows consistent decline, indicating effective optimization without overfitting.
Validation Accuracy Curve: Steadily increases across epochs, stabilizing above **77%**, confirming strong generalization.

The pretrained model establishes a new benchmark for performance in this study, highlighting the effectiveness of transfer learning in emotion recognition tasks. Its robust results across all emotional categories confirm that leveraging pretrained audio representations is a powerful strategy, particularly when dataset size or class imbalance limits training from scratch.



In the confusion matrix, all emotions are classified with high precision, *neutral* and *anger* are particularly well recognized, where *happiness* and *sadness* are also robustly predicted, higher than in any previous model. Even though our models were trained without class balancing, in minority or ambiguous classes, the pre-trained model shows higher emotion classification performance.





The per-class performance shows that the pretrained model not only achieves strong overall accuracy, but also demonstrates consistent performance across emotional classes, with no single emotion drastically underperforming. This supports the model's suitability for real-world emotion recognition, where balanced class-wise reliability is essential.

5.5 Comparison between Audio Models

The table below presents a comparative evaluation of multiple deep learning models trained on two prominent audio feature sets: **MFCC** and **Log-Mel Spectrogram**. The performance is assessed using Accuracy, Unweighted Average Recall (UAR), and F1-score.

Table 2: Results of the Audio model

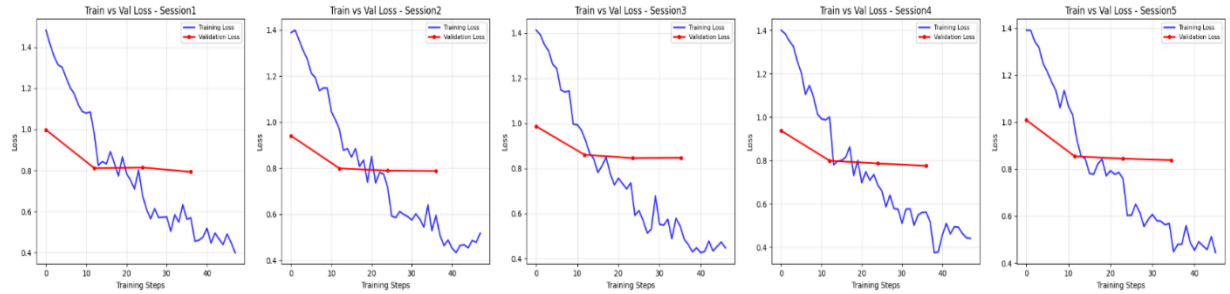
Model	MFCC			Log-Mel Spectrogram		
	Accuracy	UAR	F1-score	Accuracy	UAR	F1-score
CNN	55.06%	58.43%	53.96%	54.94%	56.74%	54.70%
LSTM	60.00%	60.75%	59.91%	58.31%	57.27%	61.68%
CNN-LSTM	54.70%	56.01%	52.88%	49.52%	52.67%	47.92%

The **pretrained model** significantly outperforms all previous models, achieving the highest accuracy (74.94%) and UAR (76.30%).

The results clearly demonstrate that model architecture and feature selection significantly impact performance. Among the handcrafted feature-based models, MFCC with LSTM provides the best results. However, the pretrained model clearly leads overall, validating the effectiveness of transfer learning in speech emotion recognition tasks.
The CNN-LSTM hybrid shows marginal improvement in UAR and F1 on Log-Mel, but overall performance is lower compared to the standalone LSTM, possibly due to overfitting or misalignment in feature processing stages, which we need to improve.

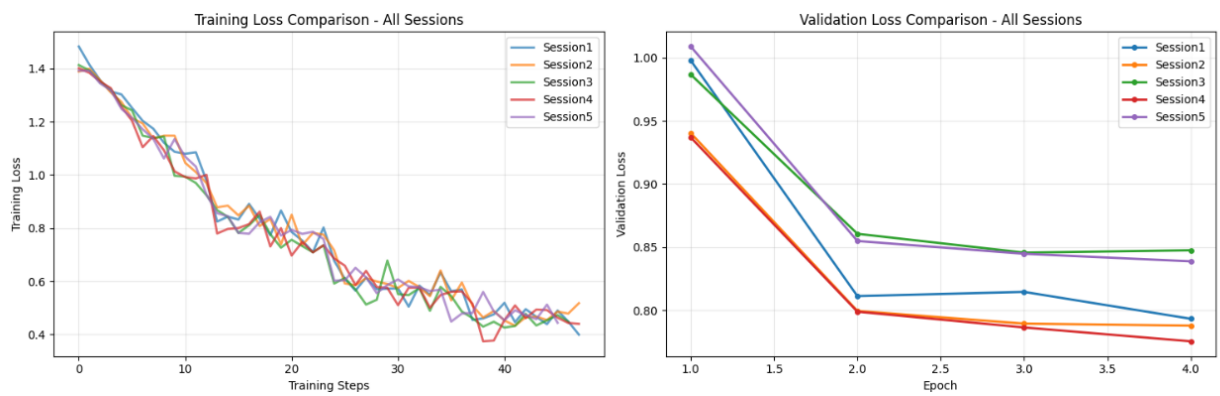
5.6 Text-only Approach:

- **Session-wise Train vs Val Loss curve:**



The training and validation loss curves across all five sessions show consistent convergence with no severe overfitting. Both losses decrease steadily, with validation loss plateauing after the second epoch in most sessions. The moderate gap between training and validation loss indicates strong generalization, attributable to the class-weighted training and session-wise cross-validation.

- **Training and Validation Loss Comparison Across Sessions:**

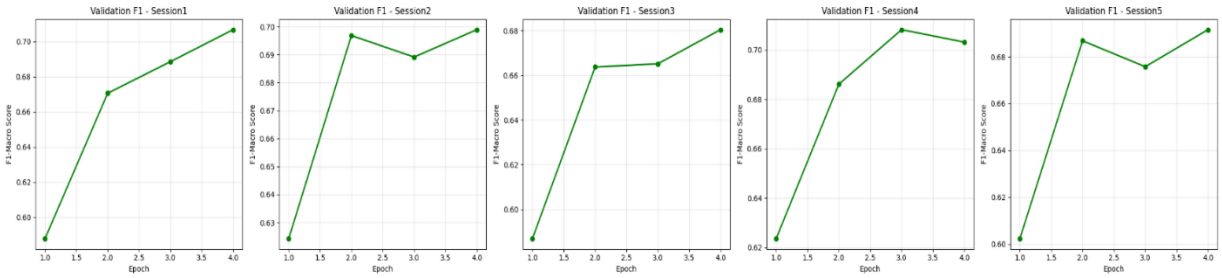


The training loss consistently decreases across all sessions, indicating smooth convergence during optimization. Validation loss follows a similar trend, with a sharp drop in the first two epochs and stabilization, thereafter, confirming that the model does not overfit and achieves balanced learning. The overlap of curves across different sessions further demonstrates robustness, suggesting that the BERT-based classifier generalizes effectively despite speaker and dialogue variation.

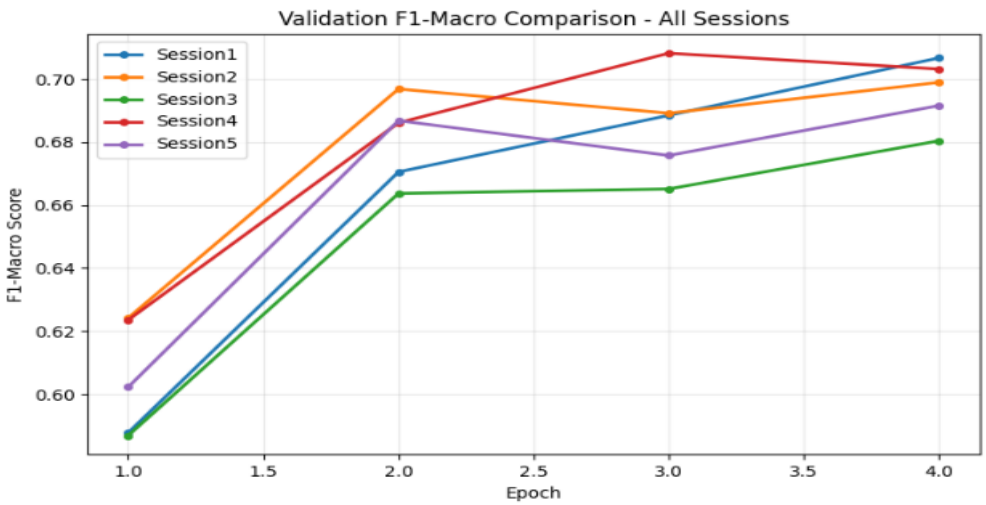
- **Session-wise Validation F1 Curves**

The validation F1 curves across all five sessions demonstrates that the model steadily improves within the first two epochs, achieving substantial gains in performance. Beyond the second epoch, the improvements become more gradual, with F1 scores stabilizing around the 0.68–0.71 range.

This trend depicted in the picture below confirms that three to four epochs are sufficient for convergence, while also highlighting consistency in generalization across different sessions and unseen speakers.



- **Session-wise Validation F1 macro curve:**

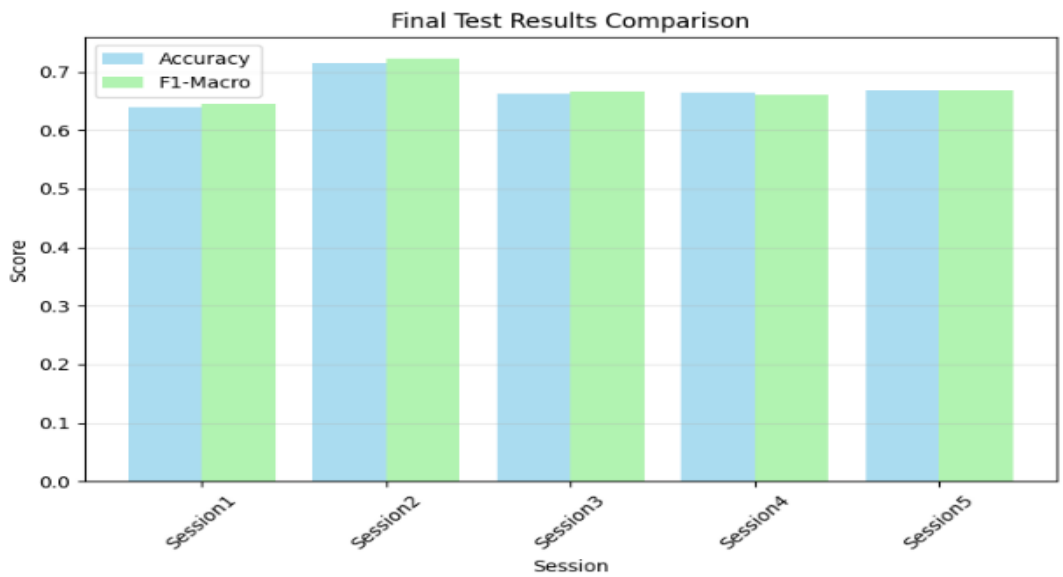


The validation F1-macro comparison reveals distinct learning flows across sessions. Sessions 2 and 4 achieved the highest peak performance, reaching approximately 0.72 F1-macro scores by epoch 4. Session 1 demonstrated the most substantial improvement, advancing from 0.59 to 0.71 across the training period. Sessions 3 and 5 showed more modest gains, with Session 3 plateauing around 0.66 and Session 5 reaching 0.68.

- **Performance result comparison session-wise:**

The final test results comparison indicates relatively consistent performance across all sessions. Session 2 achieved the best results with both accuracy and F1-macro scores reaching approximately 0.72, followed closely by Sessions 1, 4, and 5 with scores around 0.67-0.68. Session 3 showed the lowest performance at approximately 0.66 for both metrics.

The close alignment found in ***"Final Test Result Comparison"*** (given below) between accuracy and F1-macro scores across all sessions suggests balanced classification performance across emotion categories, indicating that the weighted loss function effectively addressed class imbalance issues. The consistency of results (ranging only from 0.66 to 0.72) demonstrates stable model generalization despite the leave-one-session-out cross-validation approach, with a performance variance of less than 6% across different therapy sessions



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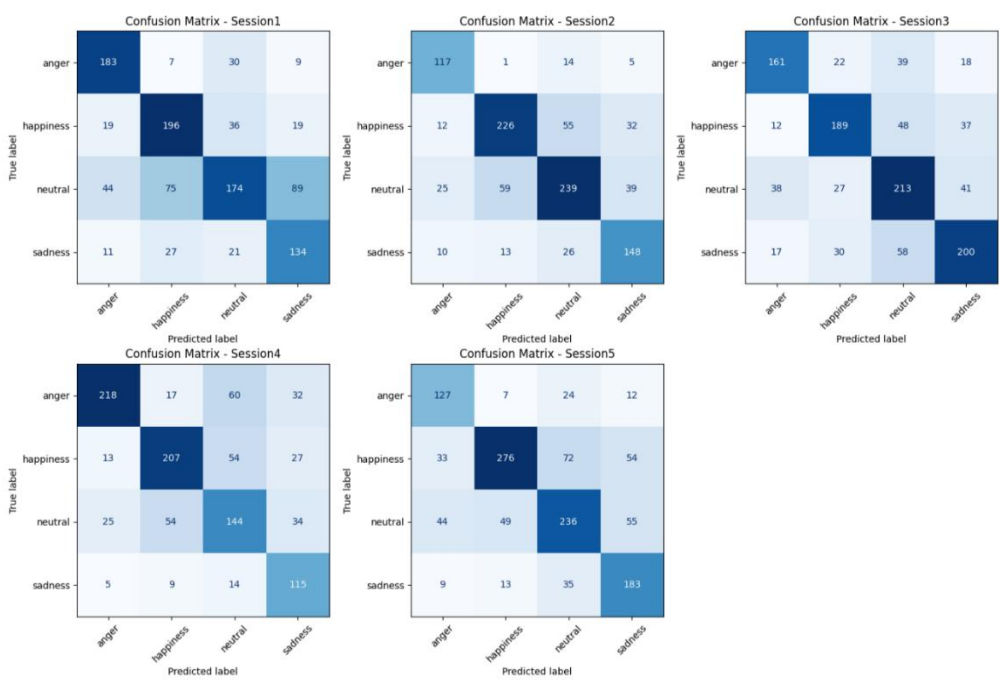
- **Session-wise Confusion matrix analysis:**



The session-wise confusion matrices reveal consistent classification patterns:
Anger consistently showed strong diagonal values across all sessions, indicating reliable detection. Happiness demonstrated good classification accuracy, but with some confusion with neutral emotions. Neutral emotions showed the highest misclassification rates, often confused with happiness and sadness. Sadness exhibited strong performance in most sessions, with particularly good results in Sessions 4 and 5.

- **Overall performance analysis:**

The BERT-based emotion classification model achieved a mean accuracy of 67.05% ± 2.74% and a mean macro-average F1-score of 67.27% ± 2.95% across all five therapy sessions. Session 2 demonstrated the best performance (71.50% accuracy, 72.29% macro F1-score), while Session 1 showed the lowest performance (63.97% accuracy, 64.49% macro F1-score).

Table 3: Overall Classification report all sessions combined

Emotion	**Precision**	**Recall**	**F1-score**	**Support**
Anger	0.72	0.73	0.72	1103
Happiness	0.73	0.68	0.70	1617
Neutral	0.63	0.59	0.61	1704
Sadness	0.61	0.72	0.66	1078

The overall classification report reveals that anger achieved the most balanced performance across precision, recall, and F1-score (72% each), while neutral emotions proved most challenging with 63% precision, 59% recall, and 61% F1-score despite being the largest class with 1,704 instances.

These findings highlight both the effectiveness of text-based emotion recognition and the necessity of integrating the audio branch into the multimodal system to capture additional emotional cues.

6. Future Work

Future work will focus on implementing multimodal fusion by combining the established audio and text branches. Three fusion strategies will be evaluated: early fusion (feature-level concatenation of MFCC/spectrograms with BERT embeddings), late fusion (decision-level combination of separate audio and text predictions), and hybrid fusion (attention-based cross-modal integration).
Additionally, a real-time deployment system will be developed with a user-friendly web interface that allows users to record speech or input text and receive immediate emotion predictions. The complete system will be comprehensively evaluated against state-of-the-art multimodal baselines, and ablation studies will be conducted to determine the most effective fusion strategy.

7. Conclusion

This research established robust unimodal baselines for speech emotion recognition using the IEMOCAP dataset. For the audio branch, the pre-trained Wav2Vec2.0 model achieved the highest accuracy of 74.94%, while a custom LSTM with MFCC features reached 60.00%

accuracy among models trained from scratch. For the text branch, the BERT-based classifier demonstrated consistent performance with an average accuracy of 67.05% ± 2.74% across session-wise cross-validation.

Key findings include the superiority of MFCC features over Log-Mel spectrograms, the effectiveness of transfer learning approaches, and the unexpected underperformance of hybrid CNN-LSTM architectures compared to standalone models. While unimodal models provided competitive performance, their limitations in capturing the full emotional context underscore the need for multimodal integration.

These findings directly reinforce the research motivation: combining audio (how something is said) and text (what is said) is expected to significantly improve emotion classification accuracy. With this foundation, the next phase of the project will focus on multimodal fusion and real-time deployment, bringing the system closer to practical, real-world applications in healthcare, call centers, education, and human-computer interaction.

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9. Appendix 1

Table 4: Addressing COs, Knowledge Profile (K), and Complex Engineering Problem (EP)

CO	K (Knowledge Profile)	EP (Complex Engineering Problem)	Model Used
CO1	**Background [K1, K2, K3]** K1: Understanding multimodal emotion recognition challenges K2: Application of deep learning for audio and text processing K3: Implementation of CNN, LSTM, and Transformer architectures.	**Background [EP1]** EP1: Requires knowledge of signal processing, NLP, and deep learning integration EP6: Cross-modal feature alignment and fusion complexities.	**CNN, LSTM, Hybrid CNN-LSTM, Wav2Vec2, BERT**
CO2	**Related Works [K8]** K8: Comparative analysis of unimodal vs multimodal approaches in emotion recognition literature	**Related Works [EP1, EP2, EP7]** EP1: Study of existing SER methodologies and their limitations EP2: Integration challenges in multimodal systems EP7: Identifying optimal fusion strategies for audio-text modalities.	**CNN-LSTM, BiLSTM with Attention, BERT-based models, Transfer Learning approaches**
CO3	**Problem Analysis [K1, K2, K3, K4]** K1: Audio feature extraction (MFCC, Log-Mel spectrograms) K2: Text preprocessing and BERT embeddings K3: Temporal sequence modeling with LSTMs K4: Multimodal fusion techniques and evaluation metrics.	**Problem Analysis [EP1, EP2, EP3, EP6, EP7]** EP1: Handling class imbalance in IEMOCAP dataset EP2: Speaker-independent emotion recognition EP3: Real-time processing constraints EP6: Cross-validation strategies for multimodal data	**Custom CNN with Channel Attention, BiLSTM with Temporal Attention, Wav2Vec2 fine-tuning**

		EP7: Generalization across different emotional expressions.	
CO4	**Design & Implementation [K5]** K5: Designing multimodal architecture with early/late fusion strategies, implementing attention mechanisms, and creating user-friendly deployment interface.	**Design & Implementation [EP1, EP2, EP3, EP6, EP7]** EP1: Architecture design for optimal feature extraction EP2: Training pipeline for multiple modalities EP3: Model optimization and hyperparameter tuning EP6: Session-wise cross-validation implementation EP7: Real-time inference system development.	**Hybrid CNN-LSTM with Dual Attention, BERT for Sequence Classification, Pre-trained Wav2Vec2**
CO5	**Materials & Devices [K6]** K6: Implementation using PyTorch, Librosa, Hugging Face Transformers, and specialized audio processing libraries.	**Materials & Devices [EP1, EP2, EP4, EP5]** EP1: GPU-accelerated training with AMP EP2: Memory optimization for large-scale audio processing EP4: Efficient batch processing for multimodal data EP5: Model deployment and inference optimization.	**PyTorch-based CNN/LSTM implementations, Hugging Face BERT and Wav2Vec2, Custom attention mechanisms**
CO6	**Social & Environmental Impact [K7]** K7: Responsible AI deployment in emotion recognition, privacy considerations in speech/text analysis, and ethical implications of automated emotion detection.	**Social & Environmental Impact [EP2, EP5, EP6]** EP2: Privacy-preserving emotion recognition systems EP5: Reducing computational carbon footprint through efficient model design EP6: Ensuring unbiased emotion recognition across diverse populations and cultures.	**Lightweight CNN architectures, Efficient BERT fine-tuning, Energy-optimized inference models**

10 Appendix 2

10.1 Social and Environmental Impact of Engineering

Societal Impact:

The proposed multimodal Speech Emotion Recognition (SER) system has notable potential to enhance human-computer interaction in daily life. It can improve virtual assistants, call centers, and online learning platforms, making them more responsive and empathetic by accurately identifying emotions through both audio and text. In healthcare, it can be used to monitor mental

health, detect stress, and provide timely support, especially in remote or underserved areas where professional help is limited.

Health Impacts:
Emotion recognition can enhance the risk of early detection of psychological conditions like depression, anxiety, and frustration by analyzing speech aspects. The system can be incorporated into telemedicine or mental health applications to promote early Assessment and improve general well-being. This is associated with protective health care and a reduction in the potential risk of emotional or psychological disorders remaining untreated.

Safety Impacts:
The system's purpose is to be supplementary to human decision-making, not to take the place of human judgment. Disclaimers will help inform users that their predictions are available for purposes of guidance rather than to be a substitute for in-person consulting actions with professionals. This should maintain safety for people and protect the system from the possible risk that users will develop an over-reliance on automated decision processes in potentially sensitive situations of need healthcare, counseling, etc.

Legal Impacts:
As the system involves speech and text data, it needs to obey the data privacy and protection rules like GDPR or similar local alternatives. Responsible and ethical usages, including risks associated with anonymization and confidentiality of user data, must be ensured. They need to be added while making it clear that the system's predictions are just supportive tools and not legally binding assessments, also minimizing liability here

Cultural Impacts:
The expression of speech and emotion will vary across languages and cultures. Through the use of datasets like IEMOCAP and eventually a multilingual corpus, it will support diverse users. Through this provision, people can benefit from emotion-aware technologies without bias or exclusion, based on language and culture.

Environmental Impacts:
The SER system is digital through and through and diminishes the reliance on paper-based communications and other manual surveys that are typically used in psychological studies. The full usage of SERs in the context of online platforms can also reduce travel and in-person measures on pre-consultation assessments, therefore reducing associated carbon emissions. Reasonable management of cloud computing resources may also further contribute to sustainability, in terms of effective energy management and minimization of environmental impact.